GARRETT HOLLAND

Senior Infrastructure & Kubernetes Platform Engineer | Go Automation | Systems Reliability

garrett.lee.holland@gmail.com | Detroit, MI | garrettholland.com | github.com/glholland

PROFESSIONAL SUMMARY

Senior infrastructure engineer and team lead with 5+ years building and operating distributed Linux container platforms. Lead an on-premises fleet of 150+ OpenShift clusters across datacenters and production facilities worldwide; own Go-based bare-metal automation; and drive reliability, security, and systems-level troubleshooting across compute, networking, and storage. Proven record delivering tested infrastructure software, reusable platform capabilities, and developer-enablement solutions at enterprise scale.

CORE EXPERTISE

Infrastructure software: Go, Python, REST APIs, CLI design, unit/integration/end-to-end testing, Terraform, Ansible, Bash

Linux & compute: RHEL/CoreOS, bare metal, KVM, KubeVirt, Proxmox, bootc, SR-IOV, NFS/iSCSI, distributed storage

Container platforms: OpenShift/Kubernetes, multi-cluster lifecycle, GitOps, networking, storage, RBAC, Quay, Harbor, disconnected image distribution

Reliability & security: 24/7 on-call, incident response, RCA, observability, load testing, resource contention, Kyverno, Cosign, SBOMs, Trivy, RHACS, Vault

PROFESSIONAL EXPERIENCE

Kubernetes Platform Engineer - Team Lead | Ford Motor Company — **11/2025 - Present**

- Lead Ford's on-premises Kubernetes team operating 150+ OpenShift clusters across Southeast Michigan datacenters and production facilities worldwide, supporting business-critical manufacturing and enterprise workloads.
- Maintain 99.8% trailing 30-day node availability while keeping the fleet at N-1 version currency through planned upgrades, lifecycle management, and production operational discipline.
- Own development and production hardening of **iloctl**, a Go CLI replacing HPE OneView-dependent bare-metal workflows with direct iLO automation across multiple server generations.
- Designed a layered CI/CD verification suite spanning unit, integration, and end-to-end tests; pipelines build the release binary and exercise it against sidecar-hosted iLO emulators representing multiple hardware generations.
- Reproduced and resolved node-level NFS contention by designing a Kubernetes load test that saturated a single node with concurrent NetApp Trident mounts; isolated RPC-stream concurrency and validated **nconnect** and **max_session_slots** tuning without changing kernel dirty-page defaults.
- Drive multi-team production incident response from triage through root-cause analysis and remediation, coordinating engineers and support teams from Red Hat, NetApp, and Microsoft.
- Contribute across the engineering lifecycle with 774 pull requests and 328 code reviews spanning 57 repositories; mentor engineers on Kubernetes architecture, automation, testing, and operational practices.

Kubernetes Security Specialist | Ford Motor Company — **05/2025 - 11/2025**

- Served as the Kubernetes subject-matter expert in Cloud Security, shaping container security, policy enforcement, and secure software-delivery architecture for engineering stakeholders.
- Authored the enterprise artifact-registry Terraform module adopted by 488 engineering teams and contributed reusable modules supporting cluster provisioning and observability automation.
- Built reusable Tekton pipelines for SBOM generation, Cosign image signing, and vulnerability scanning; published the patterns internally for organization-wide adoption.
- Implemented policy-as-code and least-privilege controls with Kyverno, Pod Security Standards, and Kubernetes RBAC across multi-tenant OpenShift environments.
- Evaluated OpenShift Virtualization at the Kubernetes workload layer, including virt-launcher operations, SR-IOV networking, storage integration, and VM-to-Kubernetes modernization paths.

Kubernetes Platform Engineer | Ford Motor Company — **12/2022 - 05/2025**

- Operated and upgraded enterprise OpenShift infrastructure, coordinating patching, lifecycle management, availability response, and capacity needs across a large multi-cluster fleet.
- Partnered with application and platform teams to design scalable Kubernetes workloads and resolve issues spanning compute resources, RBAC, networking, storage, and delivery pipelines.
- Developed reusable Terraform modules and GitOps patterns that standardized infrastructure changes across environments and reduced one-off operational work.
- Authored the Ford Docs platform, CaaS API collections, Tekton Pipelines-as-Code guides, and Workload Identity Federation documentation used to onboard 1,700+ platform tenants.
- Founded and currently lead the Ford Kubernetes User Group; coordinate a 28-person organizer team and deliver technical sessions to 100+ engineers per event.

DevOps Engineer | Ford Motor Company — **09/2020 - 12/2022**

- Engineered CI/CD systems for Android and iOS delivery using Concourse CI and CircleCI, automating tests, builds, and deployments.
- Led a hackathon team from concept through production delivery; the resulting internal developer tool remains in active use.

SYSTEMS ENGINEERING & LEADERSHIP

- Lead through influence across platform, security, application, and vendor teams; translate architecture, reliability, security, and capacity decisions into clear operational impact.
- Operate a hands-on Proxmox/KVM and OKD lab with TrueNAS-backed NFS/iSCSI storage and Harbor; prototype virtualization, container networking, distributed storage, and bootc image-mode Linux patterns.

EDUCATION

Master of Business Administration (MBA) - Oakland University | 2024

Bachelor of Science in Information Technology - Oakland University | 2020